Homework solutions for 02YELMA
Chapter: Electrodynamics

1. The magnetic field produced by a very long straight wire carrying current I is

$$B(r) = \frac{\mu_0 I}{2\pi r}$$

where r is the perpendicular distance from the wire.

Assume the square loop has side length a, and its nearer side is at distance s from the wire.

(a) Magnetic flux through the loop
Take a thin strip of width dr and height a:

$$dA = a\,dr$$

so

$$d\Phi = B(r)\,dA$$

thus

$$d\Phi = \frac{\mu_0 I}{2\pi r}\,a\,dr$$

Integrate from $r = s$ to $r = s + a$:

$$\Phi = \int_s^{s+a} \frac{\mu_0 I a}{2\pi r}\,dr$$

Therefore,

$$\boxed{\Phi = \frac{\mu_0 I a}{2\pi} \ln\left(\frac{s+a}{s}\right)}$$

(b) Induced emf when the loop is pulled away
As the loop moves away, $s = s(t)$ changes with speed

$$\frac{ds}{dt} = v$$

Using Faraday's law,

$$\mathcal{E} = -\frac{d\Phi}{dt}$$

Differentiate:

$$\mathcal{E} = -\frac{\mu_0 I a}{2\pi}\frac{d}{dt}\ln\left(\frac{s+a}{s}\right)$$

which gives

$$\boxed{\mathcal{E} = \frac{\mu_0 I a v}{2\pi}\left(\frac{1}{s} - \frac{1}{s+a}\right)}$$

As the loop moves away, the magnetic flux decreases. By Lenz's law, the induced current tries to maintain the original magnetic field, so the induced current is

$$\boxed{\text{counterclockwise.}}$$

2. Inside the solenoid,

$$\vec{B}(t) = B_0 \cos(\omega t)\,\hat{z}$$

The loop radius is

$$r = \frac{a}{2}$$

so its area is

$$A = \pi\left(\frac{a}{2}\right)^2 = \frac{\pi a^2}{4}$$

The magnetic flux is

$$\Phi(t) = BA = B_0 \cos(\omega t)\,\frac{\pi a^2}{4}$$

The induced emf is

$$\mathcal{E} = -\frac{d\Phi}{dt} = \frac{\pi a^2 B_0 \omega}{4}\sin(\omega t)$$

Using Ohm's law,

$$I = \frac{\mathcal{E}}{R}$$

we obtain

$$\boxed{I(t) = \frac{\pi a^2 B_0 \omega}{4R}\sin(\omega t)}$$

3. The magnetic field inside a long solenoid is

$$B = \mu_0 n I$$

The flux through one turn is

$$\Phi_1 = BA = \mu_0 nI\,\pi R^2$$

The total number of turns in length l is

$$N = nl$$

Thus the total flux linkage is

$$N\Phi_1 = nl\,\mu_0 nI\pi R^2$$

Using the definition of inductance,

$$L = \frac{N\Phi_1}{I}$$

we get

$$L = \mu_0 n^2 \pi R^2 l$$

Hence, the self-inductance per unit length is

$$\boxed{\frac{L}{l} = \mu_0 n^2 \pi R^2}$$

4. Electric field of a time-dependent solenoid
   The magnetic field inside the solenoid is

$$B(t) = \mu_0 nI(t)$$

Using Faraday's law,

$$\oint \vec{E} \cdot d\vec{l} = -\frac{d\Phi_B}{dt}$$

By symmetry,

$$\vec{E} = E_\phi(s)\,\hat{\phi}$$

and therefore

$$\oint \vec{E} \cdot d\vec{l} = E(2\pi s)$$

   (a) Inside the solenoid ($s < a$)
   The magnetic flux is

$$\Phi_B = B\pi s^2$$

so

$$E(2\pi s) = -\pi s^2 \frac{dB}{dt}$$

which gives

$$\boxed{E(s) = -\frac{s}{2}\frac{dB}{dt} = -\frac{\mu_0 n s}{2}\frac{dI}{dt}}$$

The direction is

$$\boxed{\hat{\phi}}$$

(b) Outside the solenoid ($s > a$)
Only the field inside contributes:

$$\Phi_B = B\pi a^2$$

thus

$$E(2\pi s) = -\pi a^2 \frac{dB}{dt}$$

so

$$\boxed{E(s) = -\frac{a^2}{2s}\frac{dB}{dt} = -\frac{\mu_0 n a^2}{2s}\frac{dI}{dt}}$$

Again, the direction is azimuthal ($\hat{\phi}$).

5. The applied voltage is

$$V(t) = V_0 \cos(\omega t)$$

The electric field between the plates is

$$E(t) = \frac{V(t)}{d} = \frac{V_0}{d}\cos(\omega t)$$

(a) Displacement current density
Using

$$\vec{J}_d = \epsilon_0 \frac{\partial \vec{E}}{\partial t}$$

we obtain

$$\boxed{\vec{J}_d(t) = -\epsilon_0 \frac{V_0 \omega}{d}\sin(\omega t)\,\hat{z}}$$

(b) Magnetic field

Using Ampère-Maxwell law,

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 \epsilon_0 \frac{d\Phi_E}{dt}$$

and cylindrical symmetry,

$$B(2\pi r) = \mu_0 \epsilon_0 \frac{d}{dt}(E \cdot \text{area})$$

For $r < R$, the enclosed area is

$$\pi r^2$$

thus

$$B(2\pi r) = \mu_0 \epsilon_0 \pi r^2 \frac{dE}{dt}$$

which gives

$$\boxed{B(r,t) = -\frac{\mu_0 \epsilon_0 V_0 \omega}{2d} r \sin(\omega t)}$$

For $r > R$, the full plate area contributes:

$$\pi R^2$$

so

$$\boxed{B(r,t) = -\frac{\mu_0 \epsilon_0 V_0 \omega}{2d} \frac{R^2}{r} \sin(\omega t)}$$

The magnetic field direction is

$$\boxed{\hat{\phi}}$$